

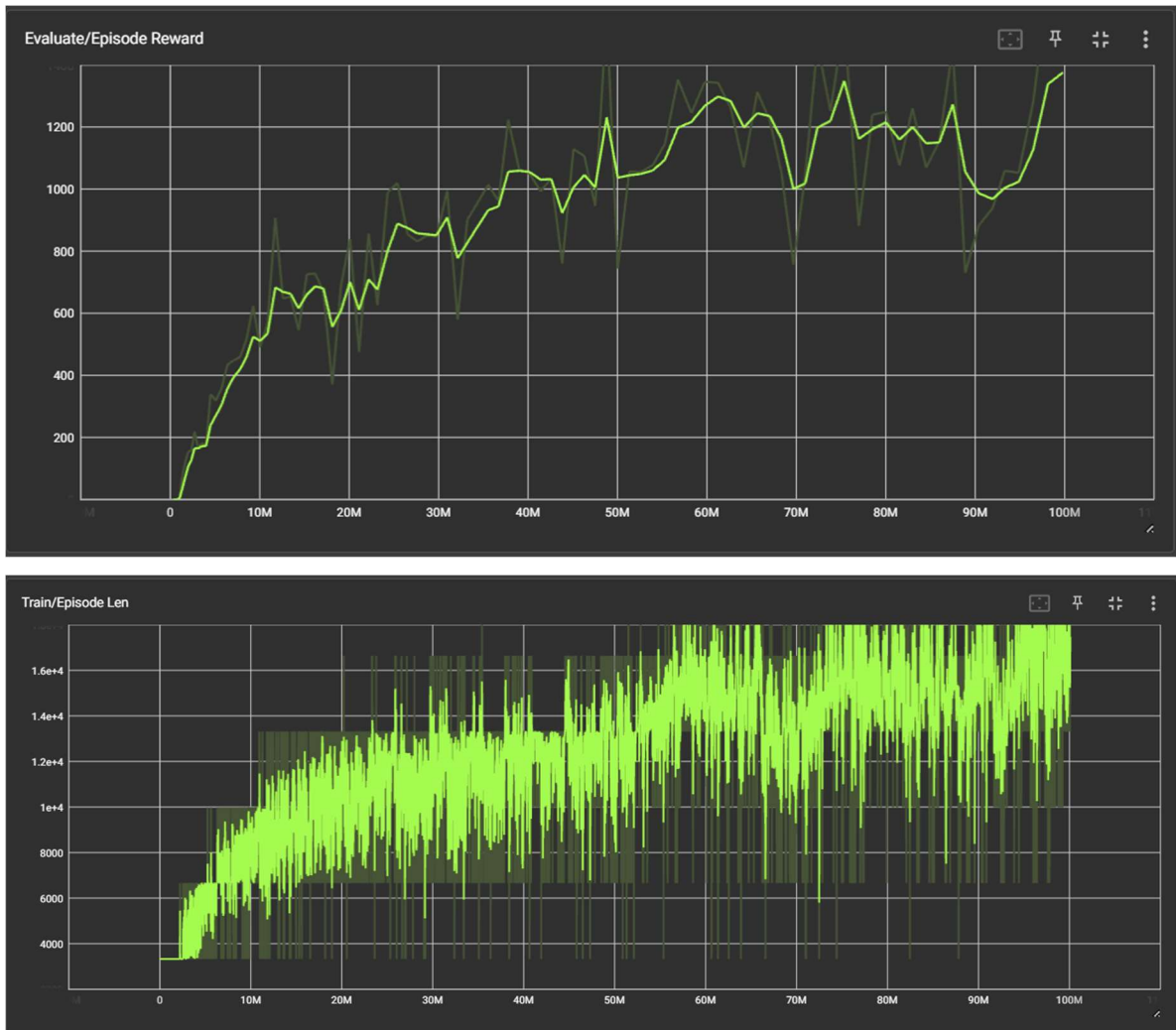
Selected topics of reinforcement learning

Lab3 – PPO

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- Screenshot of Tensorboard training curve and testing results on PPO.

- Training curve:



- Testing results (5 games):

```
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Evaluating...
/home/wesberchang/miniconda3/envs/RL/lib/python3.9/site-packages/gym/utils/passive_env_checker.py:289: UserWarning: WARN: No render fps was d
eclared in the environment (env.metadata['render_fps'] is None or not defined), rendering may occur at inconsistent fps.
  logger.warn(
/home/wesberchang/miniconda3/envs/RL/lib/python3.9/site-packages/gym/utils/passive_env_checker.py:233: DeprecationWarning: `np.bool8` is a de
precated alias for `np.bool_`. (Deprecated NumPy 1.24)
  if not isinstance(terminated, (bool, np.bool8)):
episode 1 reward: 2136.0
episode 2 reward: 1896.0
episode 3 reward: 1578.0
average score: 1870.0
=====
```

- PPO is an on-policy or an off-policy algorithm? Why? (5%)
 - PPO 為 on-policy，使用自己的最新經驗來學習。一旦策略更新了，舊的經驗就會被丟棄
 - PPO 基於當前策略（即 agent 現在使用的策略）進行更新。在訓練過程中，PPO 會使用從環境收集的“當前”策略來計算回報並更新策略。這與 off-policy 演算法（如 DQN）不同，後者則可以使用“過去”的經驗來更新策略，即使這些經驗並不是當前策略下的。

- 透過限制策略更新的幅度（使用 **confidence interval** 的概念），保持策略的穩定性和有效性，這種方法需要基於當前策略來進行學習，因此它屬於 **on-policy**。
- Explain how PPO ensures that policy updates at each step are not too large to avoid destabilization. (5%)
 - PPO 使用 **Clipped Surrogate Objective Function** 確保策略更新的程度不會過大，維持訓練的穩定性

$$L^{CLIP} = \mathbb{E}_t [\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)]$$

- ϵ is a hyper-parameter
- First term implies that the min is L^{CPI}
- Second term modifies the surrogate objective by clipping the probability ratio
- The final objective is a lower bound on L^{CPI}

(from the lecture PPT)

- Why is GAE-lambda used to estimate advantages in PPO instead of just one step advantages? How does it contribute to improving the policy learning process? (5%)
 - $\hat{A}_t = \delta_t + (\gamma\lambda)\delta_{t+1} + \dots + (\gamma\lambda)^{(T-t+1)} \delta_{T-1}$
 - 根據折扣因子對未來的步驟進行加權，從而提供一個平滑的優勢估算。
 - 一般的 one step advantages 會有高 **bias** 的問題，因為學習初期的估計通常不準確。
 - GAE-lambda 能夠提供更好的 **bias** 與 **variance** 平衡，提升策略學習的穩定性與效率。
- Please explain what the lambda parameter represents in GAE-lambda, and how adjusting the lambda parameter affects the training process and performance of PPO? (5%)
 - lambda 將不同步數的優勢估計進行指數加權平均，lambda=0 即一般的 TD-error，lambda=1 即蒙地卡羅，而當 lambda 介於 0~1 之間為 GAE 最常見的方法，近期的偏差鉅重通常較大，遠期的誤差權重會越來越低(指數減少)。Lambda 越大(ex: 0.97、0.99)代表越相信長期的 reward(低偏差、高方差)，較低的 lambda(ex: 0.9、0.95)則依賴價值函數的估計、穩定更新(高偏差、低方差)。